

Why are you here?!? - A Motivating Example

MECH 412 - *System Dynamics and Control*

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- ▶ How does “adaptive cruise control” work?
- ▶ Let the desired velocity as a function of time, the **reference** or **command**, be given by

$$r(t) = \bar{r} (1 - \exp(-at)) , \quad \forall t \geq 0,$$

where $\bar{r} \in \mathbb{R}$ and $0 < a < \infty$.

- ▶ *Problem statement:* given a desired velocity profile, have the car's velocity follow the desired velocity profile using the engine and the speedometer.

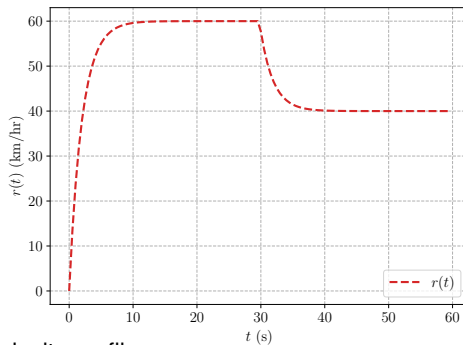
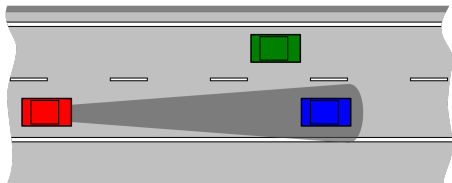


Figure 1: Desired velocity profile.

Input-Output Relationship

- ▶ First, what is the **input**, the **output (measurement)**, and the **relationship between the input and the output**.
- ▶ **Input**: force (due to the engine torque + transmission), $u(t) = f(t)$.
- ▶ **Output**: velocity (measured by a speedometer), $y(t) = v(t)$.
- ▶ The **relationship** between the **input** and the **output** is given by Newton's 2nd Law (N2L) and the measurement equation,

$$m\dot{v}(t) = \sum_{i=1}^N f_i(t),$$
$$y(t) = v(t).$$

- ▶ If drag is included, then N2L is written as

$$m\dot{v}(t) = f(t) - bv(t), \quad \forall t \geq 0,$$
$$y(t) = v(t),$$

or

$$m\dot{v}(t) + bv(t) = u(t), \quad \forall t \geq 0,$$
$$y(t) = v(t),$$

where $0 < b < \infty$.

Problem Statement Revisited

- ▶ *Problem statement revisited:* given $r(t)$, have $\lim_{t \rightarrow \infty} v(t) = r(t)$ using only $u(t)$ and $y(t)$.
- ▶ Recall,
 - ▶ **reference/command:** the desired velocity as a function of time, $r(t)$,
 - ▶ **input:** force (due to the engine torque + transmission), $u(t) = f(t)$,
 - ▶ **output:** velocity (measured by a speedometer), $y(t) = v(t)$.

Feedforward Control

- Consider the *feedforward control law*

$$u(t) = m\dot{r}(t) + br(t), \quad \forall t \geq 0.$$

- Then,

$$m\dot{v}(t) + bv(t) = u(t),$$

$$m\dot{v}(t) + bv(t) = m\dot{r}(t) + br(t),$$

$$m(\dot{r}(t) - \dot{v}(t)) + b(r(t) - v(t)) = 0,$$

$$m\dot{e}(t) + be(t) = 0, \quad \forall t \geq 0,$$

where $e(t) = r(t) - v(t)$.

- This is a constant coefficient, homogeneous, ordinary differential equation (ODE).
- To find a solution to this ODE, let

$$e(t) = \bar{e} \exp(\lambda t), \quad \dot{e}(t) = \bar{e} \lambda \exp(\lambda t).$$

► Then,

$$\begin{aligned}m\dot{e}(t) + be(t) &= 0, \\m\bar{e}\lambda \exp(\lambda t) + b\bar{e} \exp(\lambda t) &= 0, \\(m\lambda + b)\bar{e} \exp(\lambda t) &= 0.\end{aligned}$$

► Thus, $\lambda = -\frac{b}{m}$ and

$$e(t) = \bar{e} \exp\left(-\frac{b}{m}t\right), \quad \forall t \geq 0.$$

► The constant $\bar{e}$ can be found from the initial condition (IC),

$$\begin{aligned}e(0) &= \bar{e} \exp\left(-\frac{b}{m}0\right) = \bar{e}, \\ \bar{e} &= r(0) - v(0).\end{aligned}$$

► Therefore, the solution to the ODE $m\dot{e}(t) + be(t) = 0, \forall t \geq 0$, is

$$e(t) = e(0) \exp\left(-\frac{b}{m}t\right), \quad \forall t \geq 0.$$

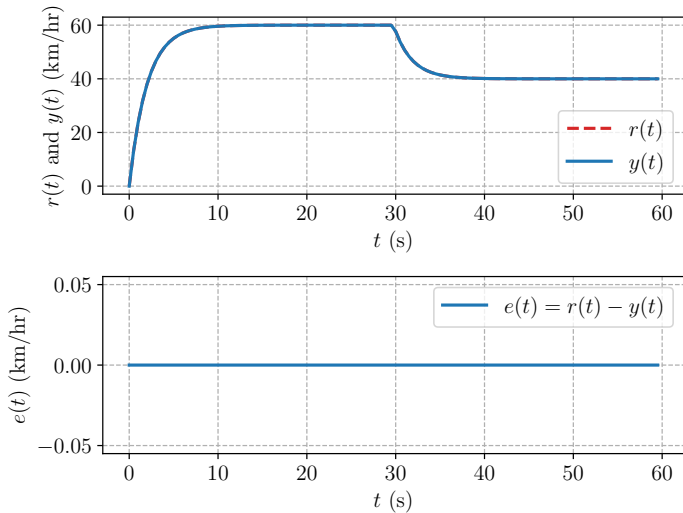


Figure 2: Velocity profile using feedforward control (no model uncertainty).

Clearly $\lim_{t \rightarrow \infty} e(t) = 0$, meaning $\lim_{t \rightarrow \infty} v(t) = r(t)$. Great!

Discussion

- ▶ The proposed feedforward control law

$$u(t) = m\dot{r}(t) + br(t), \quad \forall t \geq 0,$$

assumes

- ▶ m , the mass of the car, and
- ▶ b , the drag coefficient,

are known *exactly*, meaning there's no **model uncertainty**.

- ▶ Is “no model uncertainty” realistic? When in life do you ever know something exactly ...
- ▶ Let m_a and b_a be the *actual* (i.e., *true*) mass and drag coefficient. Apply the proposed feedforward control law and see that

$$\begin{aligned} m_a \dot{v}(t) + b_a v(t) &= u(t) \\ &= m\dot{r}(t) + br(t), \quad \forall t \geq 0, \end{aligned}$$

and ... and ... that's it ...

- ▶ Can no longer factor out m and b !

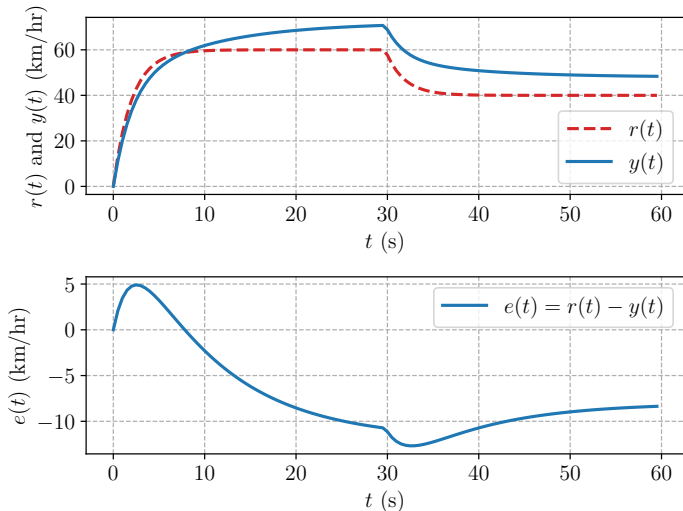


Figure 3: Velocity profile using feedforward control (with model uncertainty).

Clearly $\lim_{t \rightarrow \infty} e(t) \neq 0$, meaning $\lim_{t \rightarrow \infty} v(t) \neq r(t)$. Boo!

If m and b are not known exactly (and they are *never* known exactly) there will always be error when using a feedforward controller.

Feedback Control

- ▶ Consider the *feedback control law*

$$\begin{aligned}u(t) &= k_p(r(t) - y(t)) + k_i \int_0^t (r(\tau) - y(\tau)) d\tau \\&= k_p e(t) + k_i \int_0^t e(\tau) d\tau, \quad \forall t \geq 0,\end{aligned}$$

where k_p and k_i are constant control parameters (“control gains”) to be tuned/designed, $y(t) = v(t)$ and $e(t) = r(t) - y(t) = r(t) - v(t)$.

- ▶ This is a *proportional-integral* (PI) feedback control law.
 - ▶ The first term in the PI law is proportional to the error, $e(t)$.
 - ▶ The second term in the PI law is proportional to the integral of the error, $\int_0^t e(\tau) d\tau$.
- ▶ Then,

$$m\dot{v}(t) + bv(t) = u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau, \quad \forall t \geq 0.$$

- ▶ Yikes! How is a solution to this ODE found?
- ▶ It's actually not that hard, we just need to remember a little MATH 263. For now, just simulate.

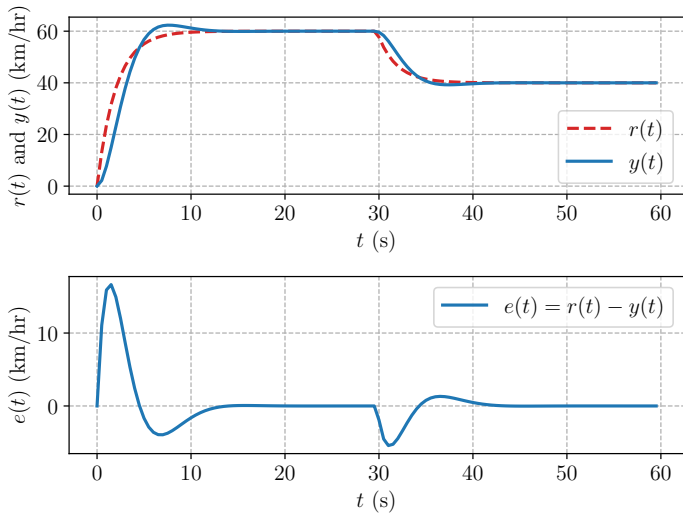


Figure 4: Velocity profile using feedback control (no model uncertainty).

Clearly $\lim_{t \rightarrow \infty} e(t) = 0$, meaning $\lim_{t \rightarrow \infty} v(t) = r(t)$. Great!

Discussion

- The proposed feedback control law

$$\begin{aligned}u(t) &= k_p(r(t) - y(t)) + k_i \int_0^t (r(\tau) - y(\tau)) d\tau \\&= k_p e(t) + k_i \int_0^t e(\tau) d\tau, \quad \forall t \geq 0,\end{aligned}$$

uses **the measurement** $y(t)$ and does not rely on knowledge of m and b .

- Let m_a and b_a be the *actual* (i.e., *true*) mass and drag coefficient. Apply the proposed control and see that

$$\begin{aligned}m_a \dot{v}(t) + b_a v(t) &= u(t) \\&= k_p(r(t) - y(t)) + k_i \int_0^t (r(\tau) - y(\tau)) d\tau \\&= k_p e(t) + k_i \int_0^t e(\tau) d\tau, \quad \forall t \geq 0,\end{aligned}$$

and ... and ... that's it!

- There's no knowledge of m and b used in the control law!

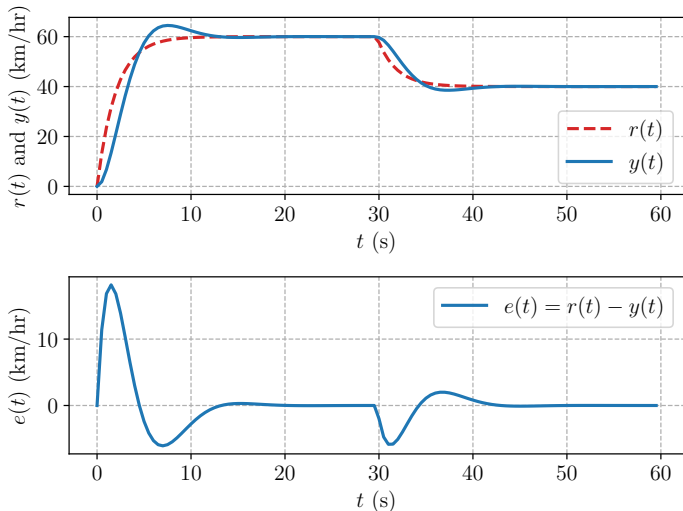


Figure 5: Velocity profile using feedback control (with model uncertainty).

Clearly $\lim_{t \rightarrow \infty} e(t) = 0$, meaning $\lim_{t \rightarrow \infty} v(t) = r(t)$, even in the presence of **model uncertainty**.
In fact, the response barely changes in the presence of model uncertainty!

The Feedforward + Feedback Control

- ▶ Comparing Figures 3 and 5,
 - ▶ the error when using feedforward control is not too large over the first few seconds, but is larger as time increases,
 - ▶ the error when using feedback control is large in the first few seconds, but goes to zero as time increases.
- ▶ Consider the use of both *feedforward* and *feedback* control applied to the *actual* (i.e., *true*) car,

$$m_a \dot{v}(t) + b_a v(t) = u(t) = \underbrace{m \dot{r}(t) + b r(t)}_{\text{feedforward}} + \underbrace{k_p e(t) + k_i \int_0^t e(\tau) d\tau}_{\text{feedback}}, \quad \forall t \geq 0,$$

where k_p and k_i are constant control parameters (“control gains”) to be tuned/designed, $y(t) = v(t)$ and $e(t) = r(t) - y(t) = r(t) - v(t)$.

- ▶ Yikes again! How is a solution to this ODE found?
- ▶ Again, to find an analytic solution to this ODE, need to remember a little MATH 263. For now, just simulate.

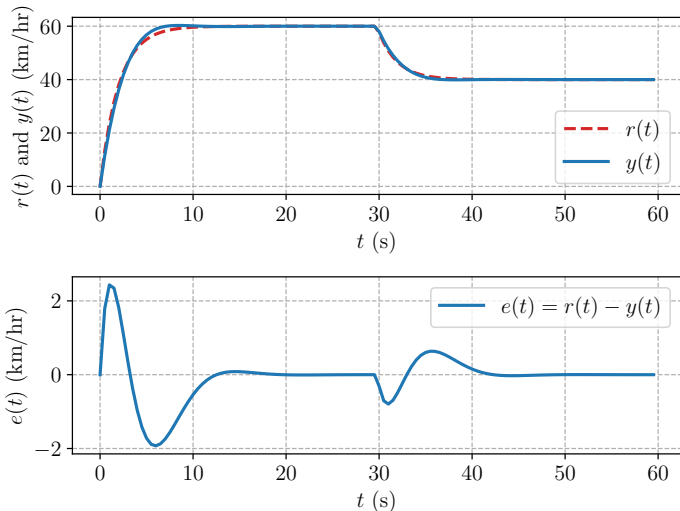


Figure 6: Velocity profile using feedforward and feedback control (with model uncertainty).

$\lim_{t \rightarrow \infty} e(t) = 0$, meaning $\lim_{t \rightarrow \infty} v(t) = r(t)$. Great!

Notice the error is much smaller than when using just feedforward or just feedback!

Discussion

- ▶ In practice, feedforward control is still useful and is often used in collaboration with feedback control.
- ▶ Feedforward control uses (imperfect) model information and the reference to anticipate what control inputs are needed to track the reference.
 - ▶ Feedforward control is *anticipatory*.
- ▶ Feedback control uses sensor information to compute control inputs that correct errors between the reference and actual response.
 - ▶ Feedforward control is *reactionary*.

“Stabilization” Example

- Consider

$$\begin{aligned}m\dot{v}(t) - bv(t) &= u(t), \quad \forall t \geq 0, \\y(t) &= v(t),\end{aligned}$$

where $0 < b < \infty$, and the control objective is to have $\lim_{t \rightarrow \infty} v(t) = 0$.

- Feedforward control won't work. Why?
- Consider the feedback control law

$$\begin{aligned}u(t) &= -k_p y(t) \\&= -k_p v(t), \quad \forall t \geq 0,\end{aligned}$$

where k_p is a constant control parameter (“control gain”) to be tuned/designed, and $y(t) = v(t)$ is the measurement.

- Again, notice that the feedback control law is not a function of m nor b . Rather, it's just a function of k_p , which *you design*.
- Then,

$$\begin{aligned}m\dot{v}(t) - bv(t) &= u(t) = -k_p v(t), \\m\dot{v}(t) + (k_p - b)v(t) &= 0, \quad \forall t \geq 0.\end{aligned}$$

► Let

$$v(t) = \bar{v} \exp(\lambda t), \quad \dot{v}(t) = \bar{v} \lambda \exp(\lambda t).$$

► Then,

$$\begin{aligned} m\dot{v}(t) + (k_p - b)v(t) &= 0, \\ m\bar{v}\lambda \exp(\lambda t) + (k_p - b)\bar{v} \exp(\lambda t) &= 0, \\ (m\lambda + (k_p - b))\bar{v} \exp(\lambda t) &= 0. \end{aligned}$$

► Thus, $\lambda = -\frac{(k_p - b)}{m}$ and

$$v(t) = \bar{v} \exp\left(-\frac{(k_p - b)}{m}t\right), \quad \forall t \geq 0.$$

- ▶ The constant $\bar{v}$ can be found from the initial condition (IC),

$$v(0) = \bar{v} \exp\left(-\frac{(k_p - b)}{m}0\right) = \bar{v}.$$

- ▶ Therefore, the solution is

$$v(t) = v(0) \exp\left(-\frac{(k_p - b)}{m}t\right), \quad \forall t \geq 0$$

- ▶ Picking k_p such that

$$k_p > b$$

leads to $\lim_{t \rightarrow \infty} v(t) = 0$!

- ▶ Note that when $k_p = 0$, meaning *no* feedback control, the response from a nonzero IC “blows up”, that is, $\lim_{t \rightarrow \infty} v(t) = \infty$.

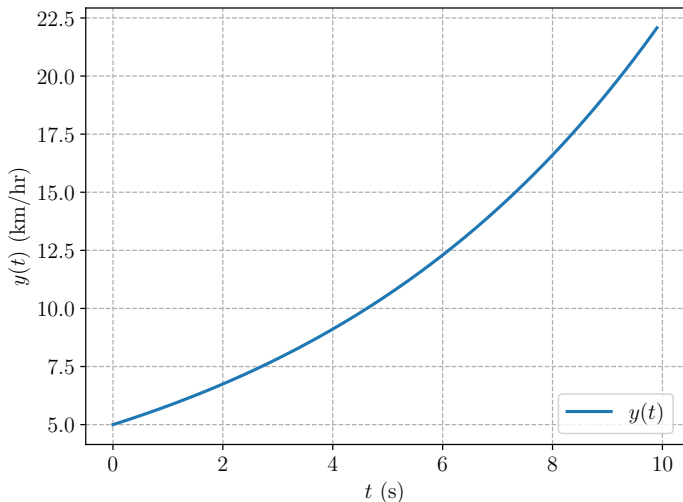


Figure 7: Response with *no* feedback control.

When there's *no* feedback control the response form a nonzero IC “blows up”, that is, $\lim_{t \rightarrow \infty} v(t) = \infty$.

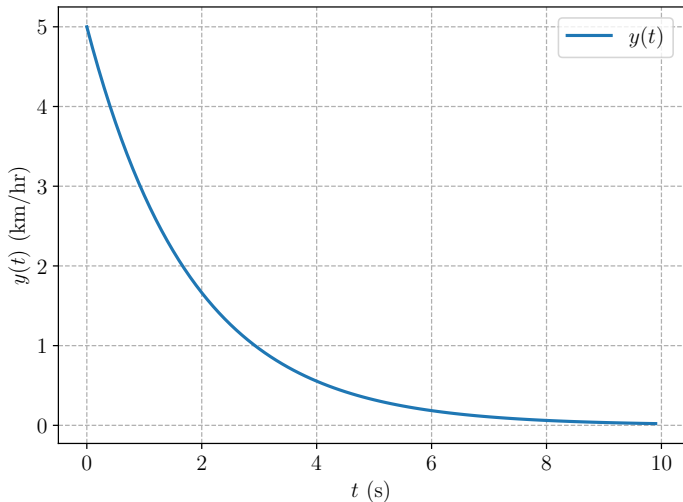


Figure 8: Response with control (no model uncertainty).

The response from a nonzero IC decays to zero, that is, $\lim_{t \rightarrow \infty} v(t) = 0$.

The naturally unstable system has been “stabilized” using simple P control.

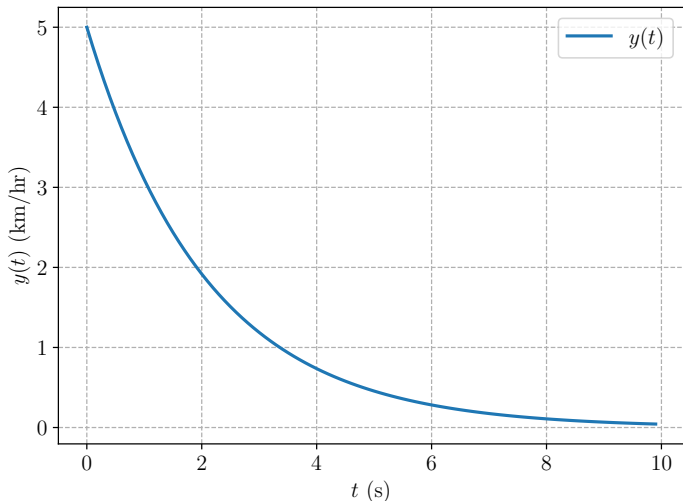


Figure 9: Response with control (with model uncertainty).

The response from a nonzero IC, in the presence of **model uncertainty**, decays to zero, that is, $\lim_{t \rightarrow \infty} v(t) = 0$.

The naturally unstable system has been “stabilized” using simple P control.

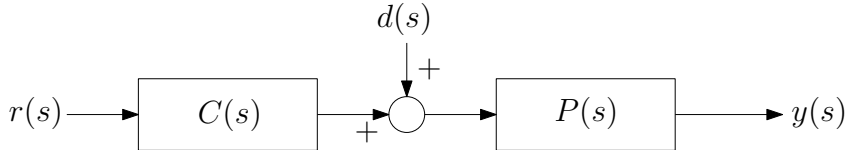


Figure 10: Feedforward control.

Feedforward control

- ▶ relies on exact model information, and
- ▶ does not use measurement $y(t)$.

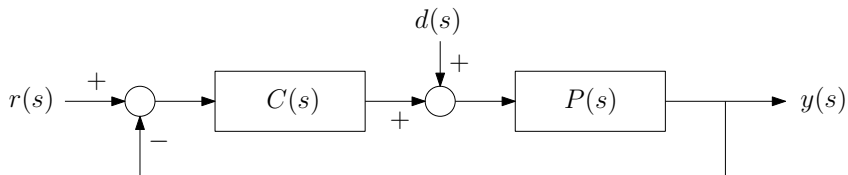


Figure 11: Feedback control.

Feedback control

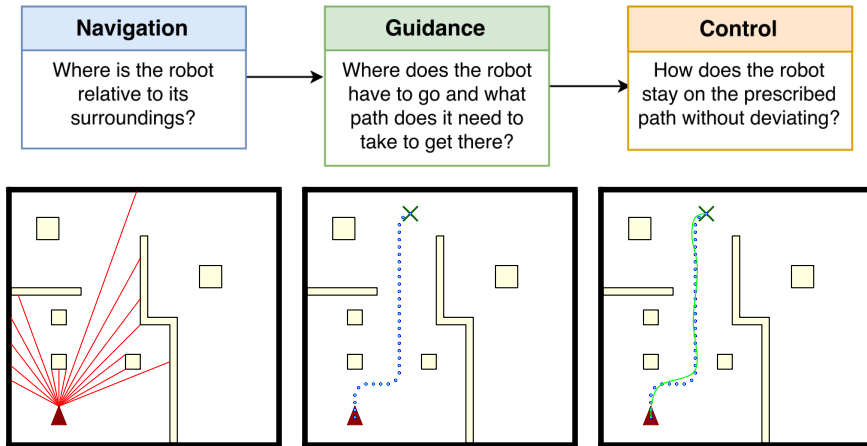
- ▶ does not rely on exact model information, and
- ▶ uses measurement $y(t)$.

What can feedback control do?

- ▶ Realize tracking of a reference (a command).
- ▶ “Stabilize” a nominally unstable system.
- ▶ Minimize the impact of disturbances and noise (i.e., reject disturbances and noise).
- ▶ Can do all of the above in the presence of model uncertainty.
- ▶ Feedback control is a **technology enabler**.

- ▶ So, **why are you here?**
- ▶ You are here to learn about feedback control systems. In particular,
 - ▶ how to derive a model that is useful for control design,
 - ▶ how to describe system characteristics in the time and frequency domain,
 - ▶ how to characterize “stability”,
 - ▶ how to design controllers,
 - ▶ how to assess robustness,
 - ▶ and more ...

The Bigger Picture - Navigation, Guidance, and Control (NGC)



- ▶ The focus of this course is the control block.
- ▶ However, solving each problem reliably is the key to making autonomous systems (e.g., a car, a plane, a robot, a chemical process, etc.) a reality.

References

Notes roughly based on [1] and the example found at <http://ctms.engin.umich.edu/CTMS/index.php?example=CruiseControl§ion=SystemModeling>.

- [1] L. Qiu and K. Zhou, *Introduction to Feedback Control*. Upper Saddle River, NJ: Prentice-Hall, Inc., 2010.